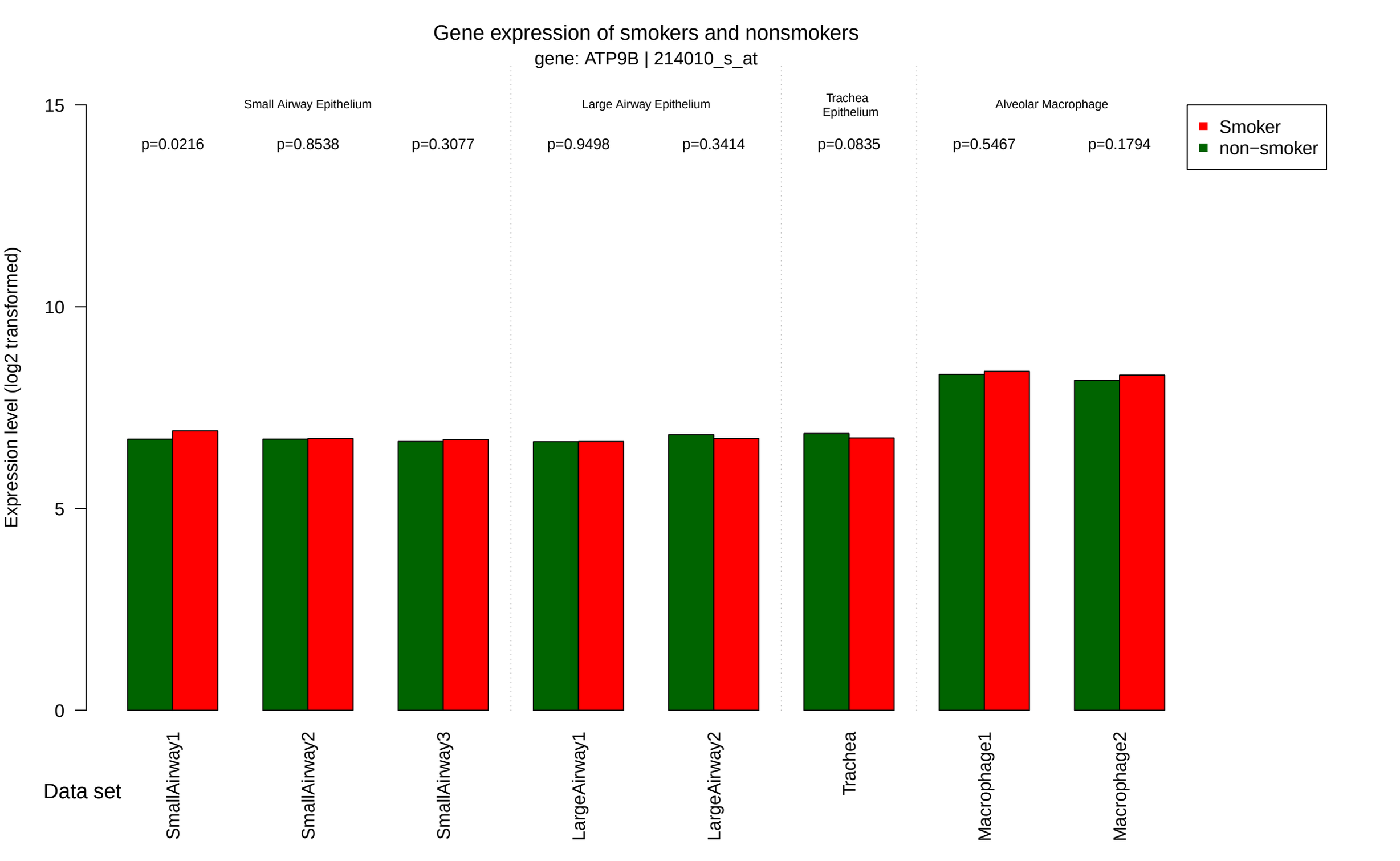
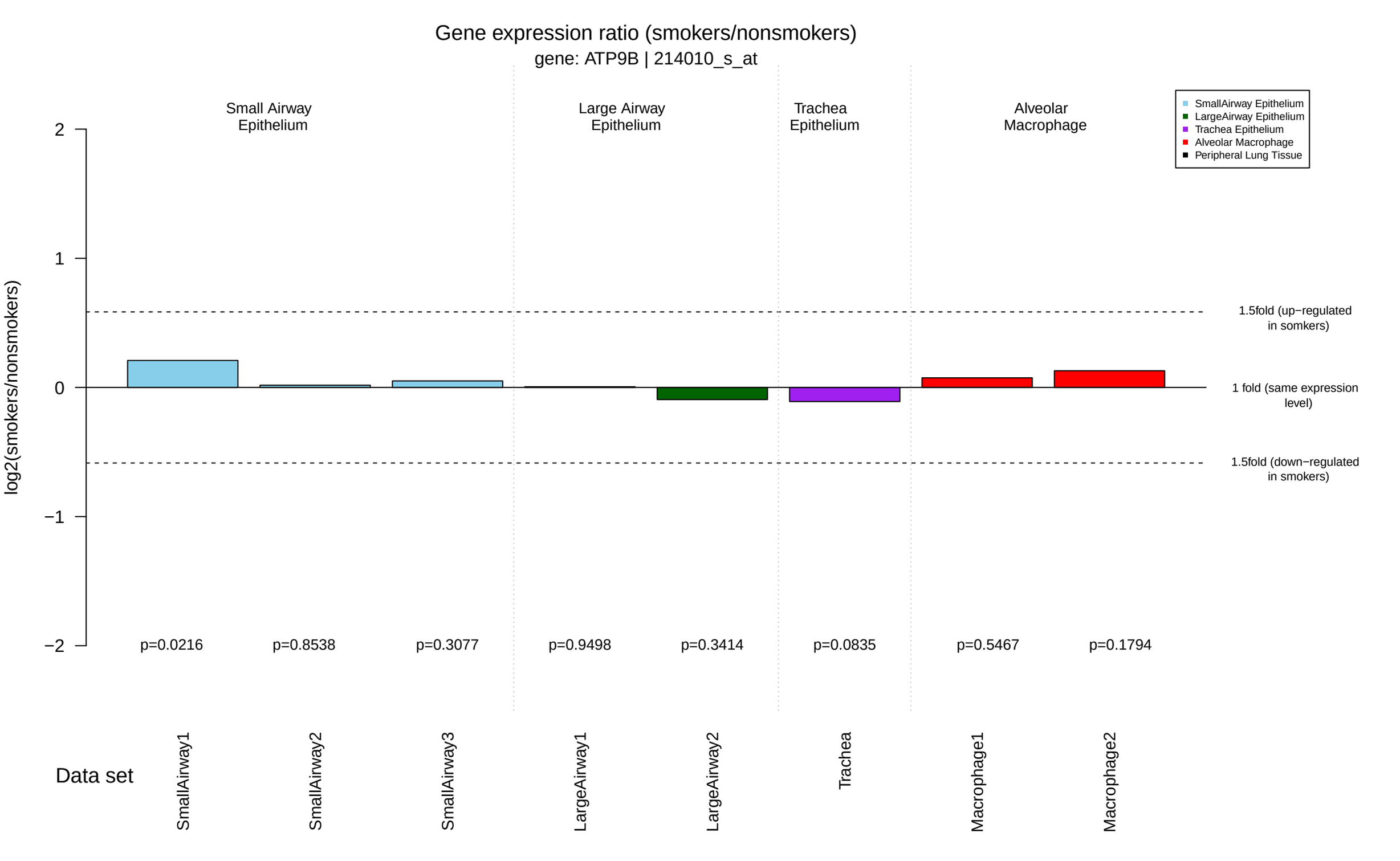
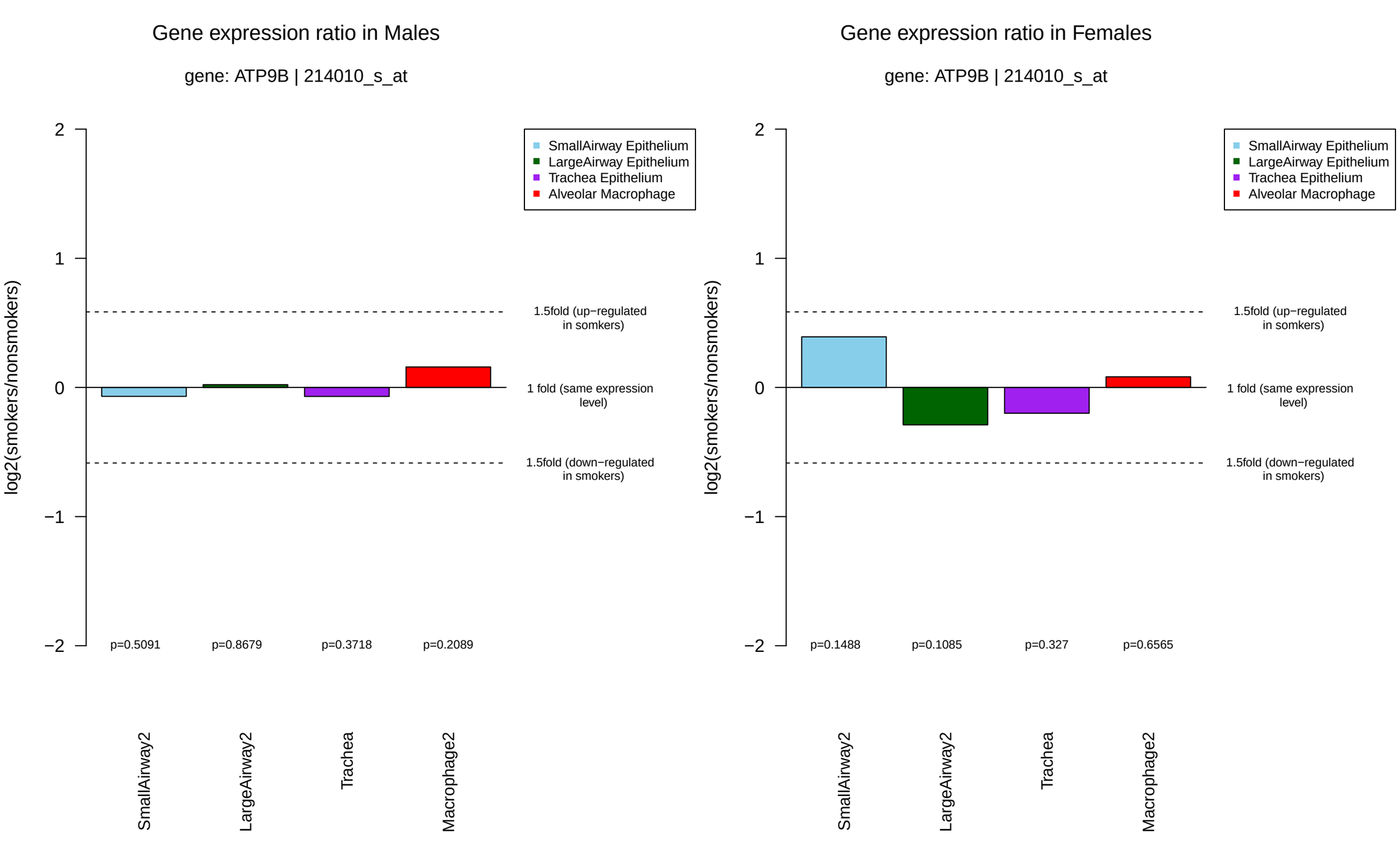
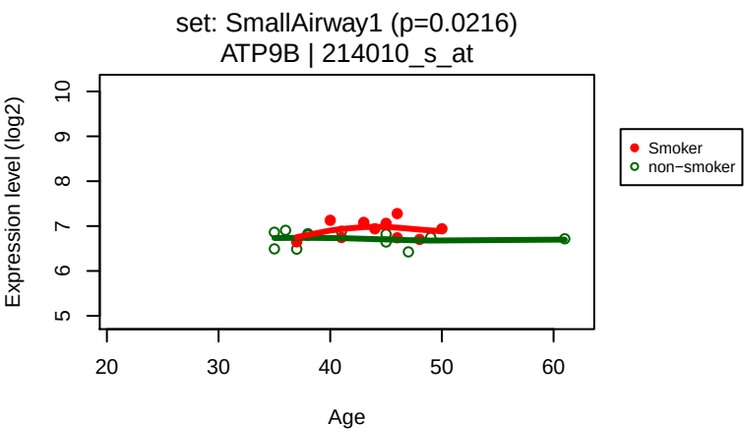
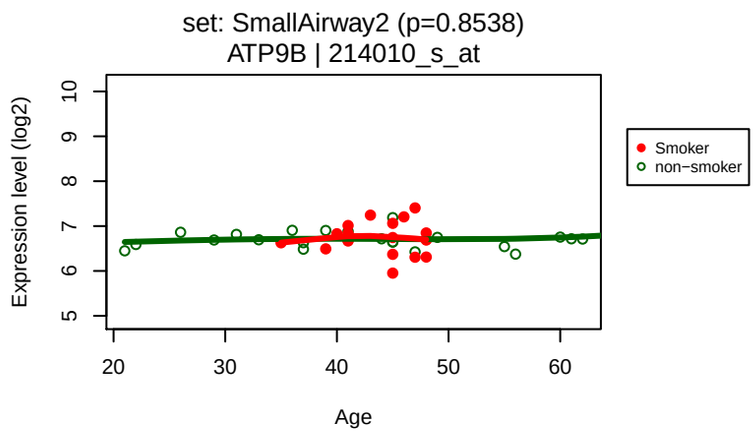
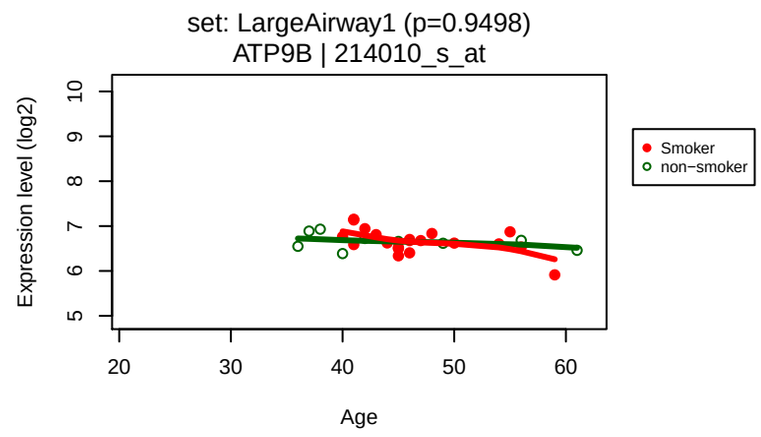

Gene expression of smokers and nonsmokers
gene: ATP9B | 214010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0216
p=0.8538
p=0.3077
p=0.9498
p=0.3414
p=0.0835
p=0.5467
p=0.1794
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

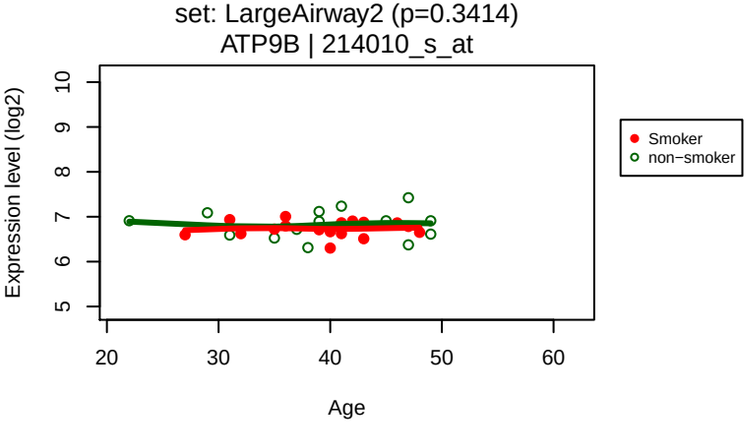

Gene expression ratio (smokers/nonsmokers)
gene: ATP9B | 214010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0216
p=0.8538
p=0.3077
p=0.9498
p=0.3414
p=0.0835
p=0.5467
p=0.1794
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

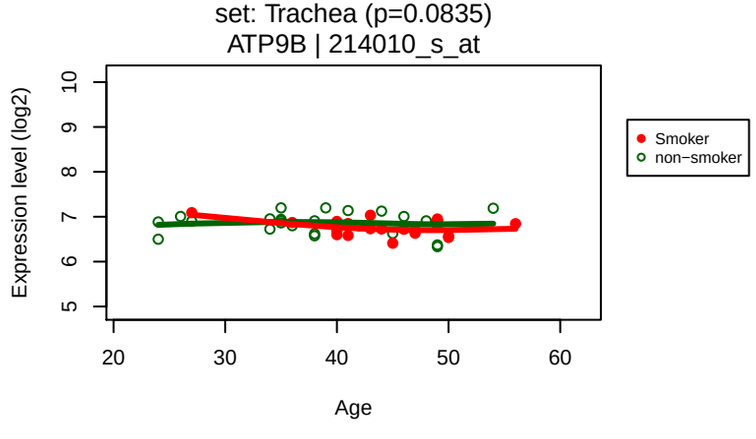

Gene expression ratio in Males
gene: ATP9B | 214010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5091
p=0.8679
p=0.3718
p=0.2089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP9B | 214010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1488
p=0.1085
p=0.327
p=0.6565
SmallAirway2
LargeAirway2
Trachea
Macrophage2

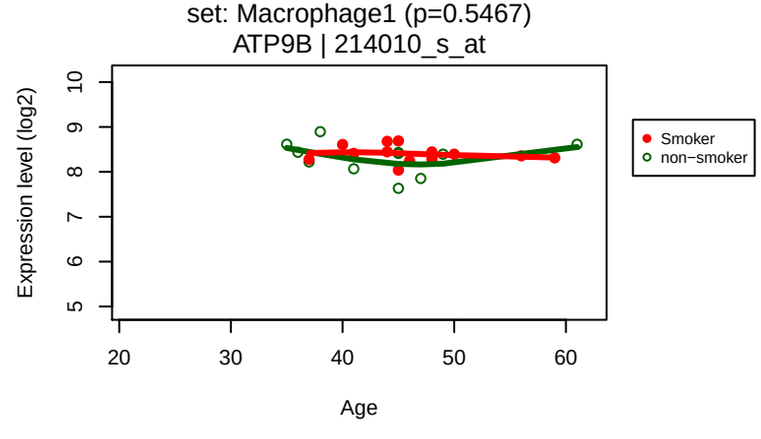

set: SmallAirway1 (p=0.0216)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker

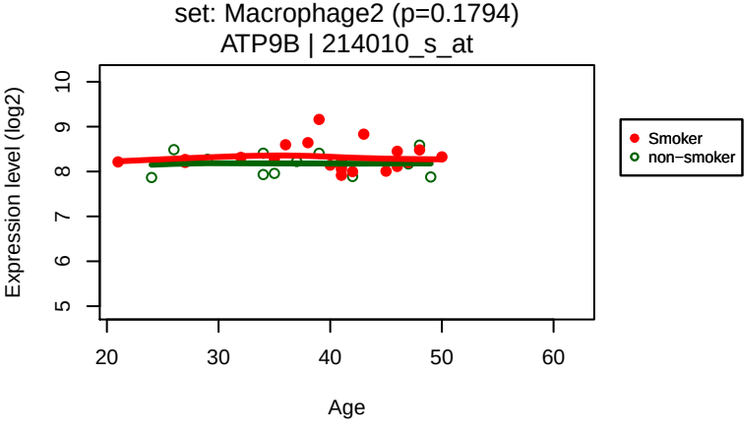

set: SmallAirway2 (p=0.8538)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker

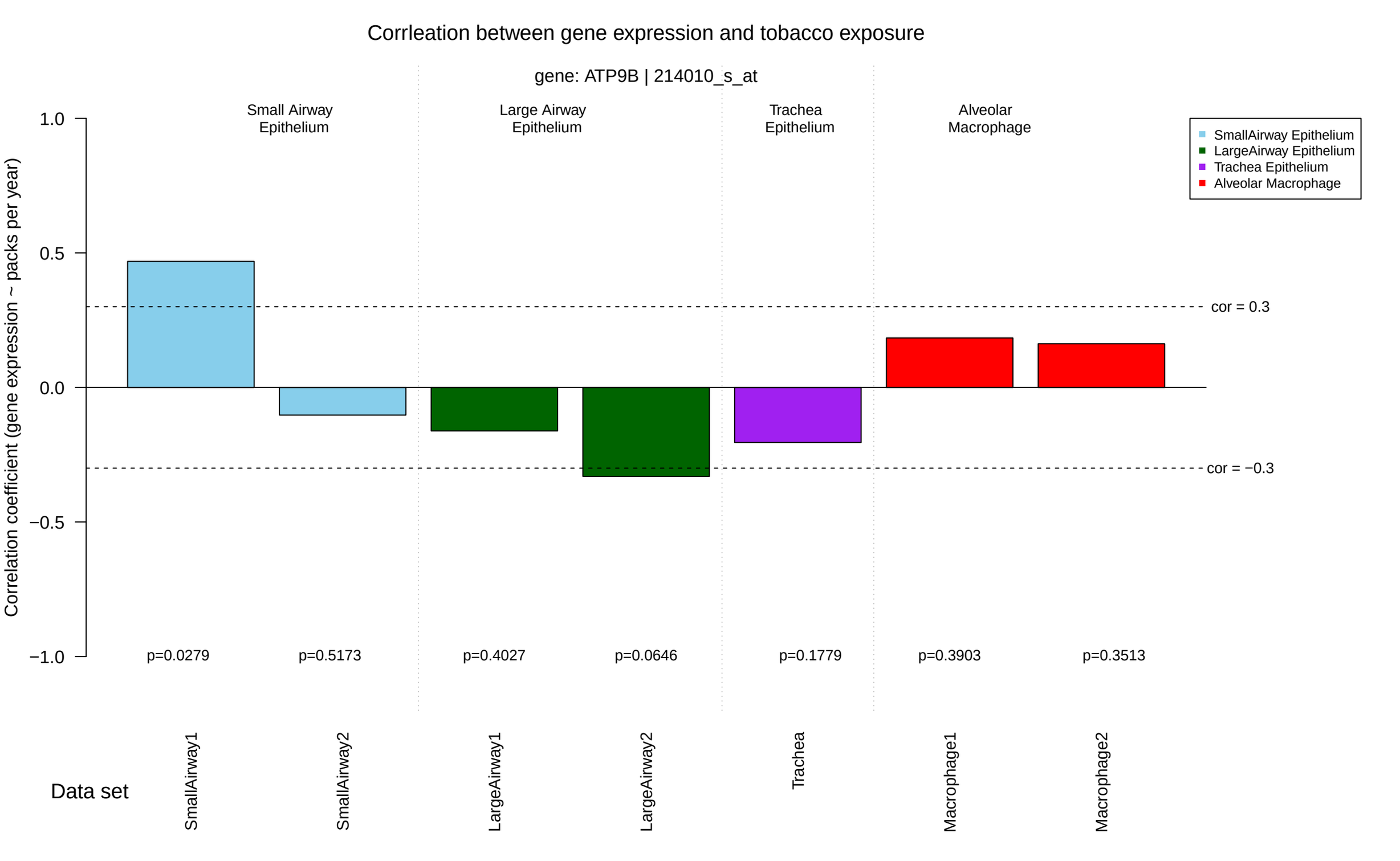

set: LargeAirway1 (p=0.9498)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3414)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0835)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5467)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1794)
ATP9B | 214010_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP9B | 214010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0279
p=0.5173
p=0.4027
p=0.0646
p=0.1779
p=0.3903
p=0.3513
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2